

2-D Diffusion Assignment (AE 703)

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

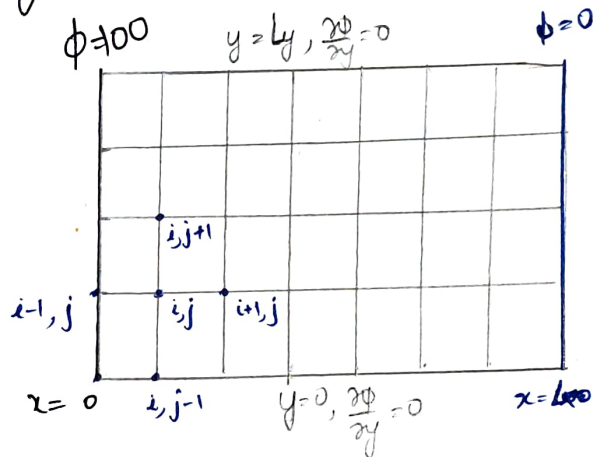
- $\phi = 100$ at $x=0$
- $\phi = 0$ at $x=L_x$
- Insulated boundaries ($\frac{\partial \phi}{\partial y} = 0$) at $y=0$ & $y=L_y$

Finite Difference Method.

In a 2D system

$$\frac{\partial^2 \phi}{\partial x^2} \approx \frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{(\Delta x)^2}$$

$$\frac{\partial^2 \phi}{\partial y^2} \approx \frac{\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}}{(\Delta y)^2}$$



As, $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

$$\frac{\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j}}{(\Delta x)^2} + \frac{\phi_{i,j-1} - 2\phi_{i,j} + \phi_{i,j+1}}{(\Delta y)^2} = 0$$

$$\beta = \frac{\Delta x}{\Delta y}$$

$$\phi_{i-1,j} - 2\phi_{i,j} + \phi_{i+1,j} + \beta^2 \phi_{i,j-1} - 2\beta^2 \phi_{i,j} + \beta^2 \phi_{i,j+1} = 0$$

$$\phi_{i-1,j} - 2(1+\beta^2)\phi_{i,j} + \phi_{i+1,j} + \beta^2 \phi_{i,j-1} + \beta^2 \phi_{i,j+1} = 0$$

At $y=0$ & $y=L_y \Rightarrow \frac{\partial \phi}{\partial y} = 0$

$$\frac{\phi_{j+1} - \phi_{j-1}}{2\Delta y} = 0$$

$$\phi_{j+1} = \phi_{j-1}$$

$$\phi_{i-1,j} - (2+2\beta^2)\phi_{i,j} + \phi_{i+1,j} + 2\beta^2\phi_{i,j+1} = 0$$

$$\{ \gamma = -(2+2\beta^2) \}$$

• At $i=1, j=1$.

$$\phi_{0,1} - (2+2\beta^2)\phi_{1,1} + \phi_{2,1} + 2\beta^2\phi_{1,0} = 0$$

$$100 + \gamma\phi_{1,1} + \phi_{2,1} + 0 = 0.$$

$$+\gamma\phi_{1,1} + \phi_{2,1} = -100.$$

• At $i=2, j=1$.

$$\phi_{1,1} + \gamma\phi_{2,1} + \phi_{3,1} + 2\beta^2\phi_{2,0} = 0$$

5 point stencil: $\gamma\phi_{i,j} + \phi_{i+1,j} + \phi_{i-1,j} + \beta^2(\phi_{i,j+1} + \phi_{i,j-1}) = 0$

$$\gamma = -(2+2\beta^2)$$

$$\begin{bmatrix} \gamma & 1 & 0 & 2\beta^2 & 0 & 0 & 0 & 0 & 0 \\ 1 & \gamma & 1 & 0 & 2\beta^2 & 0 & 0 & 0 & 0 \\ 0 & 1 & \gamma & 0 & 0 & 2\beta^2 & 0 & 0 & 0 \\ \beta^2 & 0 & 0 & \gamma & 1 & 0 & \beta^2 & 0 & 0 \\ 0 & \beta^2 & 0 & 1 & \gamma & 1 & 0 & \beta^2 & 0 \\ 0 & 0 & \beta^2 & 0 & 1 & \gamma & 0 & 0 & \beta^2 \\ 0 & 0 & 0 & 2\beta^2 & 0 & 0 & \gamma & 1 & 0 \\ 0 & 0 & 0 & 0 & 2\beta^2 & 0 & 1 & \gamma & 1 \\ 0 & 0 & 0 & 0 & 0 & 2\beta^2 & 0 & 1 & \gamma \end{bmatrix} \begin{bmatrix} \phi_{1,1} \\ \phi_{2,1} \\ \phi_{3,1} \\ \phi_{1,2} \\ \phi_{2,2} \\ \phi_{3,2} \\ \phi_{1,3} \\ \phi_{2,3} \\ \phi_{3,3} \end{bmatrix} = \begin{bmatrix} -\phi_{1,0}\beta^2 - 100 \\ -\phi_{2,0}\beta^2 \\ -\phi_{3,0}\beta^2 \\ -100 \\ 0 \\ 0 \\ -\phi_{1,4}\beta^2 - 100 \\ -\phi_{2,4}\beta^2 \\ -\phi_{3,4}\beta^2 \end{bmatrix}$$